

Course Syllabus

Theory of Computation

Computer Engineering, KMITL

Instructor: Assoc. Prof. Dr. Kietikul Jearanaitanakij

Email: kietikul.je@kmitl.ac.th

Course Line group :



Lecture playlist on YouTube  : Study clip(s) before attending each class.

<https://youtube.com/playlist?list=PLDDdXBbnvycQI9ENIW-uQrLZi7VSi6lF->

Textbook: “[An Introduction to Formal Languages and Automata](#)”, by Peter Linz.

Handout:

https://drive.google.com/file/d/1w5gOaLO6l6RHVW_fo4KPS7sZoRp6Mw9S/view?usp=sharing

Exercise and homework:

https://drive.google.com/file/d/1KsjTg2FTLlutqwmQp6ktVW0ax_GqVYx/view?usp=sharing

Grading:

- **Homework** (10 submissions, binary grading) = 30%

Link for homework submission:

<https://forms.gle/EVem3RgQDtGo8HxX8>

- **Assignment:** Web Application of Regular Expression
= 20%

Link for creating the assignment group:

<https://docs.google.com/spreadsheets/d/1RQH20YCWMi7UCK-LHIGfNHpfQjb-fxDNZl8dsvRNdMw/edit?usp=sharing>

Link for assignment submission:

<https://forms.gle/8g4cZ2DUm13TNbCh8>

- **Final exam** = 50% (Closed book)

Tentative Agenda

Week	Topics
1	Introduction
2	Mathematical Preliminaries
3	Deterministic Finite Automata
4	Nondeterministic Finite Automata
5	Regular Expression and Regular Grammar
6	Properties of Regular Language
7	Pumping Lemma
8	More Applications of Pumping Lemma
9	Context-free Languages
10	Context-free Grammars
11	Pushdown Automata
12	Properties of Context-free Languages
13	Pumping Lemma for Context-free Languages
14	More Applications of Pumping Lemma
15	Turing Machine